

IGCSE • Cambridge (CIE) • Biology

 47 mins  13 questions

Target Test

Created 30/05/26

Total Marks

/47

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1 (a) Fig. 1 shows the marine iguana, *Amblyrhynchus cristatus*.



Fig. 1

(i)

Marine iguanas are reptiles.

State **two** features that are used to classify animals as reptiles.

[2]

(ii)

State **two** structures that are present in plant cells that are not present in the cells of reptiles.

[2]

(4 marks)

(b) Extended Only

Marine iguanas feed on seaweed. Seaweed contains starch.

(i)

State the names of **two** parts of the digestive system where starch is digested by enzymes.

[2]

(ii)

Explain why the shape of an enzyme is important for digestion.

[3]

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(5 marks)

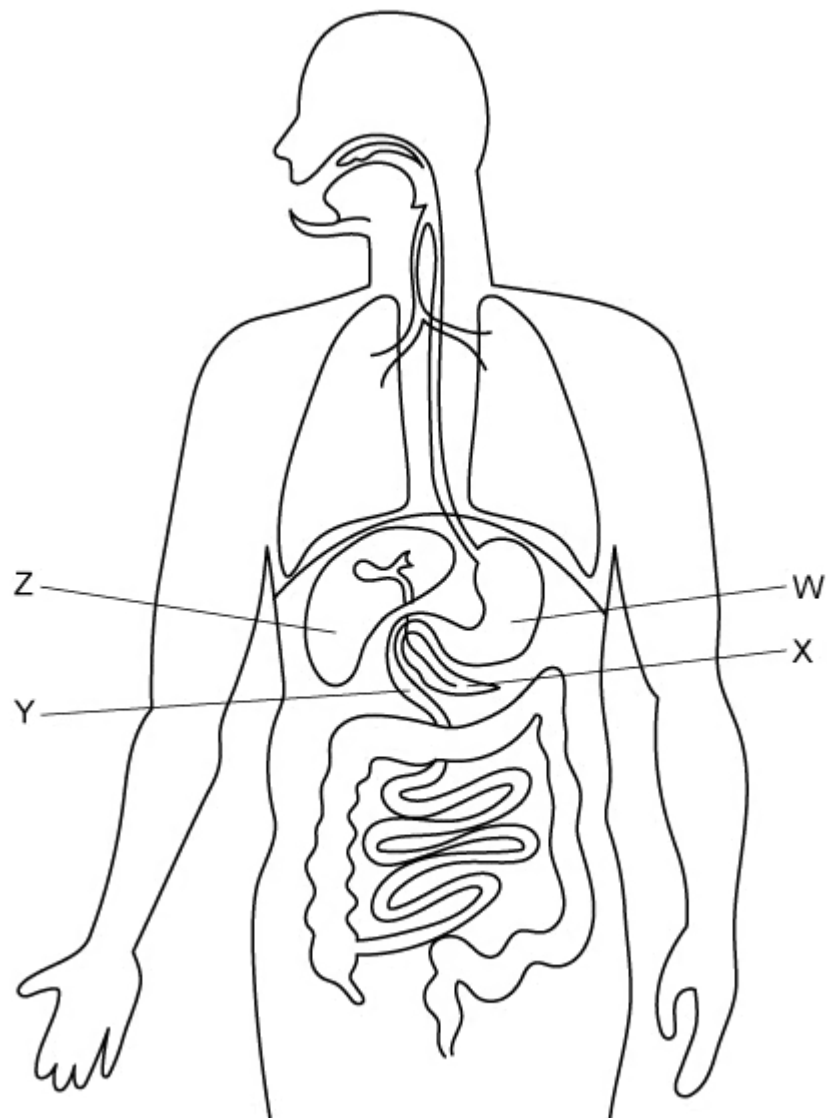
2 How does physical digestion help chemical digestion?

- A. It changes large insoluble molecules into small soluble molecules.
- B. It increases the surface area of food.
- C. It provides the acidic conditions needed.
- D. It absorbs digested nutrients directly into the blood.

(1 mark)

3 Extended only

The diagram below shows the organs of the digestive system.



In which organs does the digestion of proteins take place?

- A.** W and Y
- B.** W and Z
- C.** Z only
- D.** W and X

(1 mark)

4 Which disease can be caused by a lack of vitamin C?

- A. AIDS
- B. Coronary heart disease
- C. Rickets
- D. Scurvy

(1 mark)

5 What are the products when proteins are broken down?

- A. amino acids
- B. fatty acids
- C. glycerol
- D. simple sugars

(1 mark)

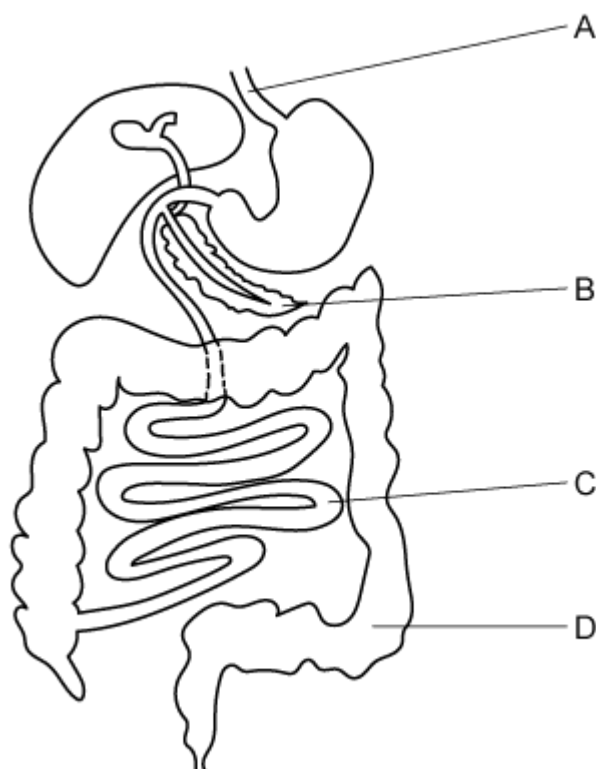
6 Which of the following is a correct cause and effect?

	deficiency	cause	effect
A	iron	diet deficient in red meat, leafy green vegetables and liver	brittle bones (rickets)
B	iron	lack of sunlight, fish and eggs	anaemia - tiredness and weakness due to not enough oxygen being delivered to tissues
C	vitamin D	lack of sunlight	brittle bones (rickets)
D	vitamin D	diet deficient in red meat, leafy green vegetables and liver	anaemia - tiredness and weakness due to not enough oxygen being delivered to tissues

(1 mark)

7 The diagram below shows part of the human digestive system.

In which structure is most of the water from food absorbed?



(1 mark)

8 A new smoothie that claims to provide a balanced diet contains the following ingredients:

Ingredients
Soya protein
Iron
Fibre
Low-fat milk
Water

What ingredients are missing from the smoothie for it to be a balanced diet product?

- A.** calcium, carbohydrates, vitamins (C and D), fats and oils
- B.** carbohydrates, vitamins (C and D), fats and oils
- C.** vitamins (C and D) and mineral ions
- D.** vitamins (C and D) and calcium

(1 mark)

- 9** Red pandas, *Ailurus fulgens*, and humans have a similar arrangement of teeth.

Fig. 1 shows a section through one tooth of a red panda. Fig. 2 shows the side view of the lower jaw of a red panda.

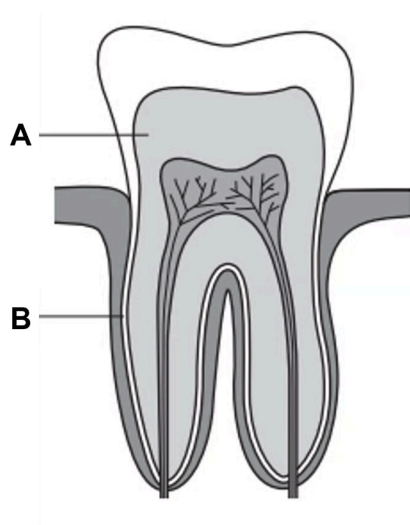


Fig. 1

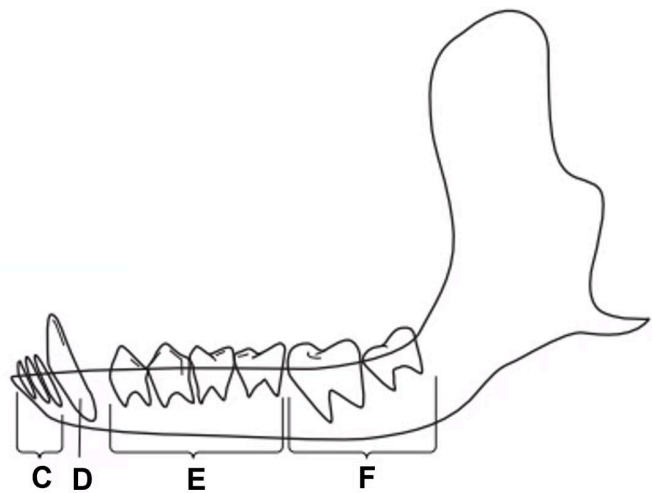


Fig. 2

(i)

State the names of the structures labelled **A** to **F** in Fig. 1 and Fig. 2.

[3]

(ii)

State the type of digestion that breaks down large pieces of food.

[1]

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(4 marks)

10 Which of the following describe prevention methods for rickets?

- A. vitamin D supplementation, adequate intake of protein, adequate intake of iron.
- B. vitamin D supplementation, adequate intake of calcium, avoid exposure to sunlight
- C. vitamin D supplementation, adequate intake of calcium, exposure to sunlight
- D. foods that are high in vitamin D, vitamin A supplementation, adequate intake of fibre

(1 mark)

11 (a) **Extended Only**

Fig. 1 shows a villus from the small intestine of a mammal and an enlarged view of a cell from region A.

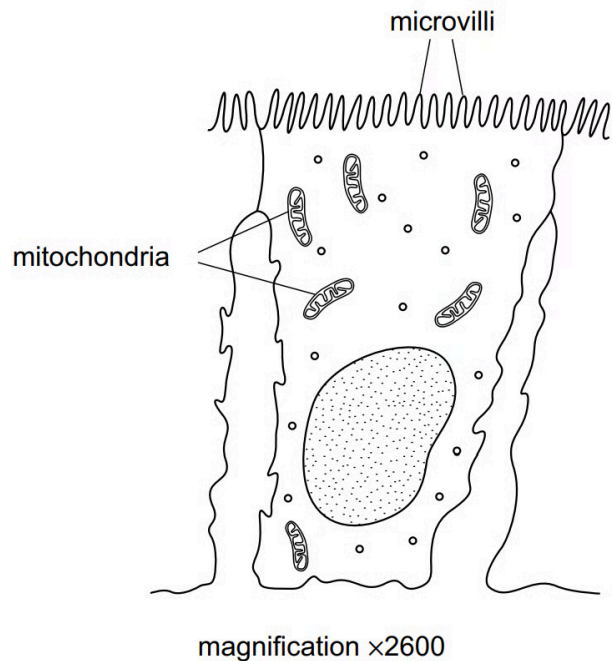
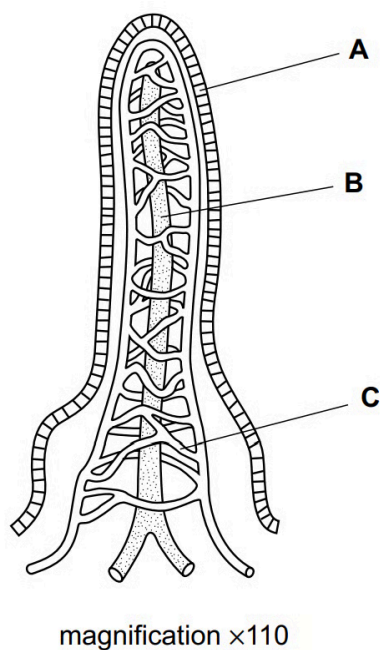


Fig. 1

Name regions A, B and C.

(3 marks)

(b) Extended Only

Explain why the cells from region **A** have many microvilli and many mitochondria.

(3 marks)

- 12 (a)** During digestion, enzymes act on different types of food to produce simpler substances that can be absorbed.

Complete Table 1 by inserting the missing information.

Table 1

food type	enzyme acting on the food type	simpler substances produced
Protein	Protease	
	Amylase	
		Fatty acids and glycerol

(5 marks)

- (b)** The following diagram includes some of the stages of food breakdown and descriptions of these stages.

Complete the following by drawing lines to connect each stage of food breakdown with the correct description.

Stage of food breakdown		Description
Ingestion		Breakdown of large insoluble molecules into small soluble ones
Absorption		Digested food molecules move into the cells of the body
Assimilation		Digested food molecules move through the wall of the intestine
Chemical digestion		Taking in food and drink through the mouth

(4 marks)

(c) Enzymes are involved with the chemical digestion of food molecules.

State **two** parts of the digestive system where chemical digestion occurs.

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(2 marks)

13 (a) Bile is required for the effective digestion of fats.

Outline the role of bile in fat digestion.

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..... **(2 marks)**

(b) In patients with cystic fibrosis, thick mucus blocks the pancreatic duct.

Explain why a blockage of the pancreatic duct leads to poor growth and 'fatty stools' (faeces containing undigested fats).

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..... **(4 marks)**

(c) Patients with cystic fibrosis often take enzyme supplement capsules with every meal.

A patient with cystic fibrosis absorbs 12 g of fat per day without enzyme capsules. When taking enzyme supplement capsules with every meal, they absorb 50 g of fat per day.

Calculate the percentage increase in fat absorption when using the capsules.

Give your answer to two significant figures.

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..... **(3 marks)**